



Lab 04: Implement Load Balancing on Compute Engine

Challenge Lab | Google Cloud Associate Cloud Engineer | GSP313

⚠ This is a Challenge Lab — no step-by-step instructions provided.
Skills from Labs 01, 02, and 03 were applied independently to complete all tasks.

Overview

Applied all load balancing skills learned in Labs 01-03 independently in a timed challenge. Created web server VMs, configured a Network Load Balancer, and set up an HTTP Application Load Balancer — all without step-by-step guidance.

Task 1 — Create Web Server Instances

Created 3 VMs (web1, web2, web3) with Apache installed and a firewall rule to allow HTTP traffic on port 80. This task combined skills from Lab 01.

Key config values used:

- Machine type: e2-small
- Image: debian-12 / debian-cloud
- Tags: network-lb-tag
- Firewall rule: www-firewall-network-lb

VM instances

Create instance

Import VM

Refresh

Instances

Observability

Instance schedules

VM instances

Filter Enter property name or value

☐	Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Co
☐	✓	web1	us-east4-c			10.150.0.5 (nic0)	34.48.178.153 (nic0)	S
☐	✓	web2	us-east4-c			10.150.0.6 (nic0)	34.86.9.75 (nic0)	S
☐	✓	web3	us-east4-c			10.150.0.7 (nic0)	35.230.166.53 (nic0)	S

Related actions

```

STATUS: RUNNING
student_01_765f3114b4d7@cloudshell:~ (qwiklabs-gcp-04-8a8656aa83d8)$ curl http://34.48.178.153
<h3>Web Server: web1</h3>
student_01_765f3114b4d7@cloudshell:~ (qwiklabs-gcp-04-8a8656aa83d8)$ curl http://34.86.9.75
<h3>Web Server: web2</h3>
student_01_765f3114b4d7@cloudshell:~ (qwiklabs-gcp-04-8a8656aa83d8)$ curl http://35.230.166.53
<h3>Web Server: web3</h3>
student_01_765f3114b4d7@cloudshell:~ (qwiklabs-gcp-04-8a8656aa83d8)$

```

Task 2 — Configure the Load Balancing Service

Set up a Network Load Balancer using a static external IP, target pool, and forwarding rule. This combined skills from Lab 01.

Key config values used:

- Static external IP: network-lb-ip-1
- Target pool: www-pool
- Port: 80

Load balancersBackendsFrontendsService LB policies

Filter

Enter property name or value

☐	Name	Load balancer type	Access type	Protocols	Region	Backends	Actions
☐	www-pool	Network (Passthrough target-pool)	External	TCP	us-east4	✔ 1 target pool (3 instances)	⋮

To view or delete load balancing resources like forwarding rules and target proxies, go to the

[load balancing components view](#).

Task 3 — Create an HTTP Load Balancer

Set up a full Application Load Balancer using an instance template, managed instance group, backend service, URL map, HTTP proxy, and forwarding rule. This combined skills from Lab 02.

Key config values used:

- Backend template: lb-backend-template
- Tags: allow-health-check
- Managed instance group: lb-backend-group
- Machine type: e2-medium
- Firewall rule: fw-allow-health-check
- External IP: lb-ipv4-1
- Health check: http-basic-check
- URL map: web-map-http
- HTTP proxy: http-lb-proxy

⋮

Load balancersBackendsFrontendsService LB policies

⋮

Filter

Enter property name or value

☐	Name	Load balancer type	Access type	Protocols	Region	Backends
☐	web-map-http	Application (Classic)	External	HTTP		✔ 1 backend service (1 instance group)
☐	www-pool	Network (Passthrough target-pool)	External	TCP	us-east4	✔ 1 target pool (3 instances)

⋮

To view or delete load balancing resources like forwarding rules and target

Backend

Backend services

1. web-backend-service

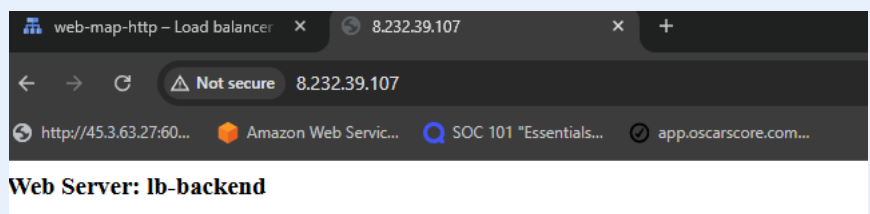
Endpoint protocol	HTTP
Named port	http
Timeout	30 seconds
Health check	http-basic-check
Cloud CDN	Disabled
Logging	Disabled
Backend security policy	None

[Show advanced](#)

Backends

! Some of your instance groups don't define the key name and value for the port name http. [Learn more](#)

Name ↑	Type	Scope	Healthy	Autoscaling	Balancing mode	Traffic duration	Selected ports ?	Capacity
lb-backend-group	Instance group	us-east4-c	✓ 2 of 2 View Details	No configuration	N/A	Default (Short)	! None	100%



What I Learned

- Challenge labs test real understanding — no instructions means you must know the commands
- NLB (Lab 01) and ALB (Lab 02) have different resource requirements and use cases
- Reading error messages carefully is key to debugging without guidance
- Resource naming matters — wrong names cause tasks to fail scoring
- Time management is critical — 25 minutes goes fast when troubleshooting

Challenges & How I Solved Them

- Created the VM instances without the startup script initially. This meant Apache was not installed and the curl verification would not work. Fixed by deleting all 3 VMs and recreating them with the startup script included in the command.

- The challenge lab outline did not include the startup script in the Task 1 commands which caused the initial mistake. Learned to always check that startup scripts are included when creating web server VMs.
- Used the wrong IP when testing with curl. The internal IP does not work for external curl tests. Fixed by using the external IP from the EXTERNAL_IP column in gcloud compute instances list.

Reflection

This challenge lab proved that I can apply load balancing concepts independently without step-by-step guidance. Completing all 3 tasks demonstrates practical knowledge of both Network and Application Load Balancers on Google Cloud.

Resources

GCP Load Balancing Docs: <https://cloud.google.com/load-balancing/docs>
Lab ID: GSP313